

Running FasterCap

Launching FasterCap from the shell

FasterCap can be launched in GUI-less mode from a System shell (e.g. a DOS shell under MS Windows, a KSH under Linux, etc.).

To enter the GUI-less text mode, you need to pass the additional command option '-b' ('batch' mode) when launching FasterCap from the System shell.

For example:

```
FasterCap -b cube.txt -a0.01
```

Please note that the command options can be passed in arbitrary order. Please note also that the default option for the GUI-less mode is not Automatic, as it is instead in the GUI mode; so you need to specify the -a option with the relevant tolerance to trigger the Automatic solution mode.

For a list of valid running command options, please see the explanation in the [Run Menu dialog box](#) paragraph. The command options are indicated together with the relevant settings descriptions. Example: **'Stop when relative error is lower than (-a)'** indicates that the relevant parameter to control this setting when launched from a System shell is '-a'

You can also get a list of the valid command options with a short description when launching FasterCap from a System shell with the '-' parameter (or with any invalid unsupported parameter).

For example:

```
FasterCap -b -?
```

```
Usage: fastercap <input file> [-a<relative error>] [-ap]
        [-m<mesh>] [-mc<mesh curvature>] [-t<tolerance>]
        [-d<interaction coeff>] [-f<outofcore>] [-g]
        [-pj] [-ps<dimension>] [-o] [-r] [-c] [-i] [-v]
```

DEFAULT VALUES:

- a: Automatically calculate settings, stop when relative error is lower than = 0.01
- ap: Automatic preconditioner usage
- m: Mesh relative refinement value = 0.1
- mc: Mesh curvature coefficient = 3
- t: GMRES iteration tolerance = 0.01
- d: Direct potential interaction coefficient to mesh refinement ratio = 1
- f: Out-Of-Core free memory to link memory condition = 5
- g: Use Galerkin scheme
- pj: Use Jacobi Preconditioner
- ps: Use two-levels preconditioner with dimension = 5

OPTIONS:

- o: Output refined geometry in FastCap2 format
- r: Dump Gmres residual at each iteration
- c: Dump charge densities in output file
- i: Dump detailed time and memory information
- v: Verbose output
- b: Launch as console/shell application without GUI
- b?: Print console usage (this text)
- bv: Print only the version
- h: Open only the help system (Linux only)

The list of valid return (exit) codes is:

```
// no error, normal end
#define FC_NORMAL_END 0
// catchall for generic error
#define FC_GENERIC_ERROR 1
// command line usage error
#define FC_COMMAND_LINE_ERROR 64
// cannot open input file(s)
#define FC_CANNOT_OPEN_FILE 66
// out of memory
#define FC_OUT_OF_MEMORY 71
// generic file error
#define FC_FILE_ERROR 74
// license error (e.g. invalid license)
#define FC_LICENSE_ERROR 77
// cannot go out-of-core (probably disk full)
#define FC_CANNOT_GO_OOC 97
```

```
// unknown exception caught
#define FC_EXCEPTION_ERROR 98
// user-requested break (not valid for console mode)
#define FC_USER_BREAK 125
```

The return codes can be checked under Linux with the following command:

```
echo $?
```

and under MS Windows with the command:

```
echo %ERRORLEVEL%
```